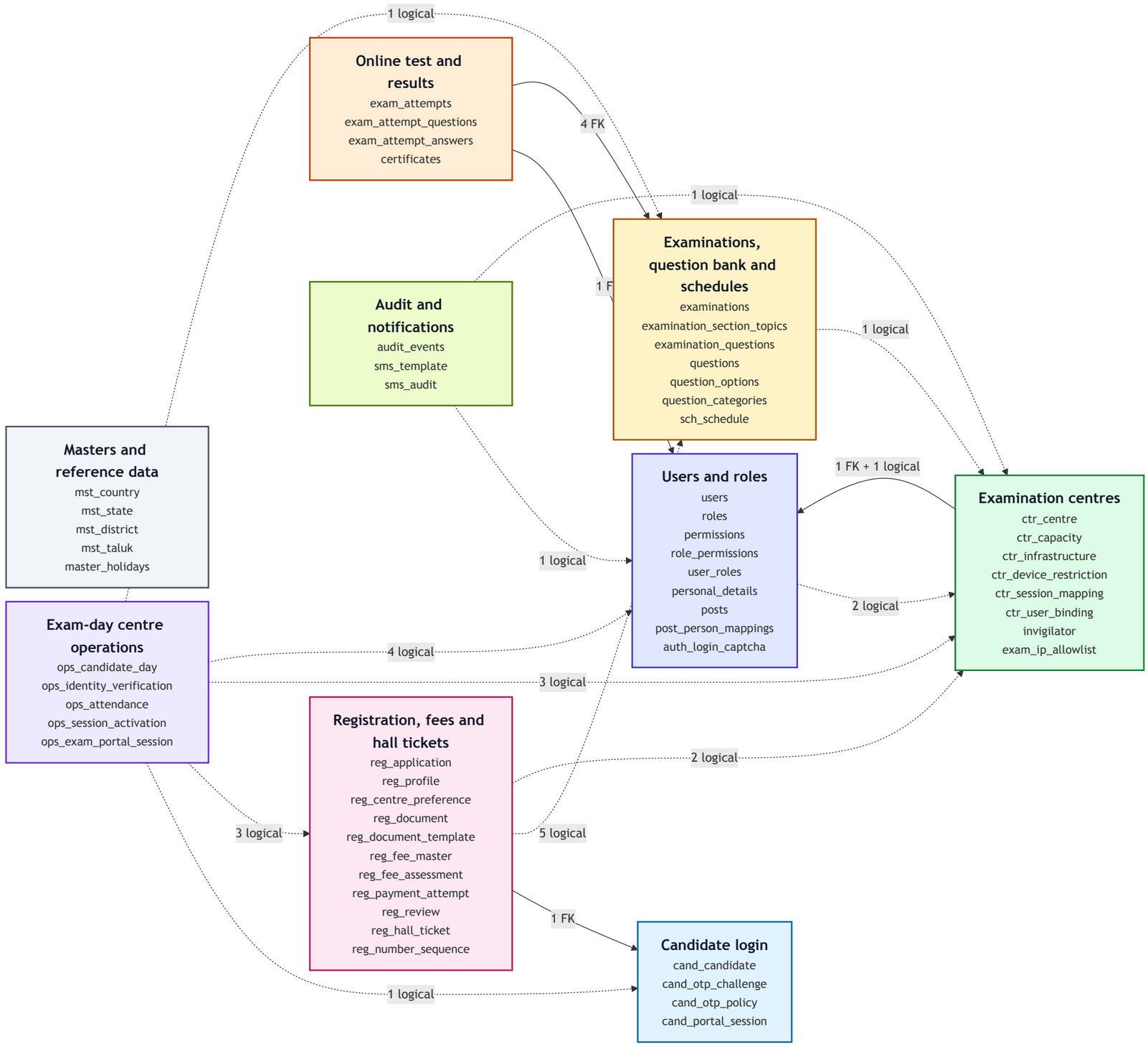
Examination Management Portal — Entity Relationship Diagram

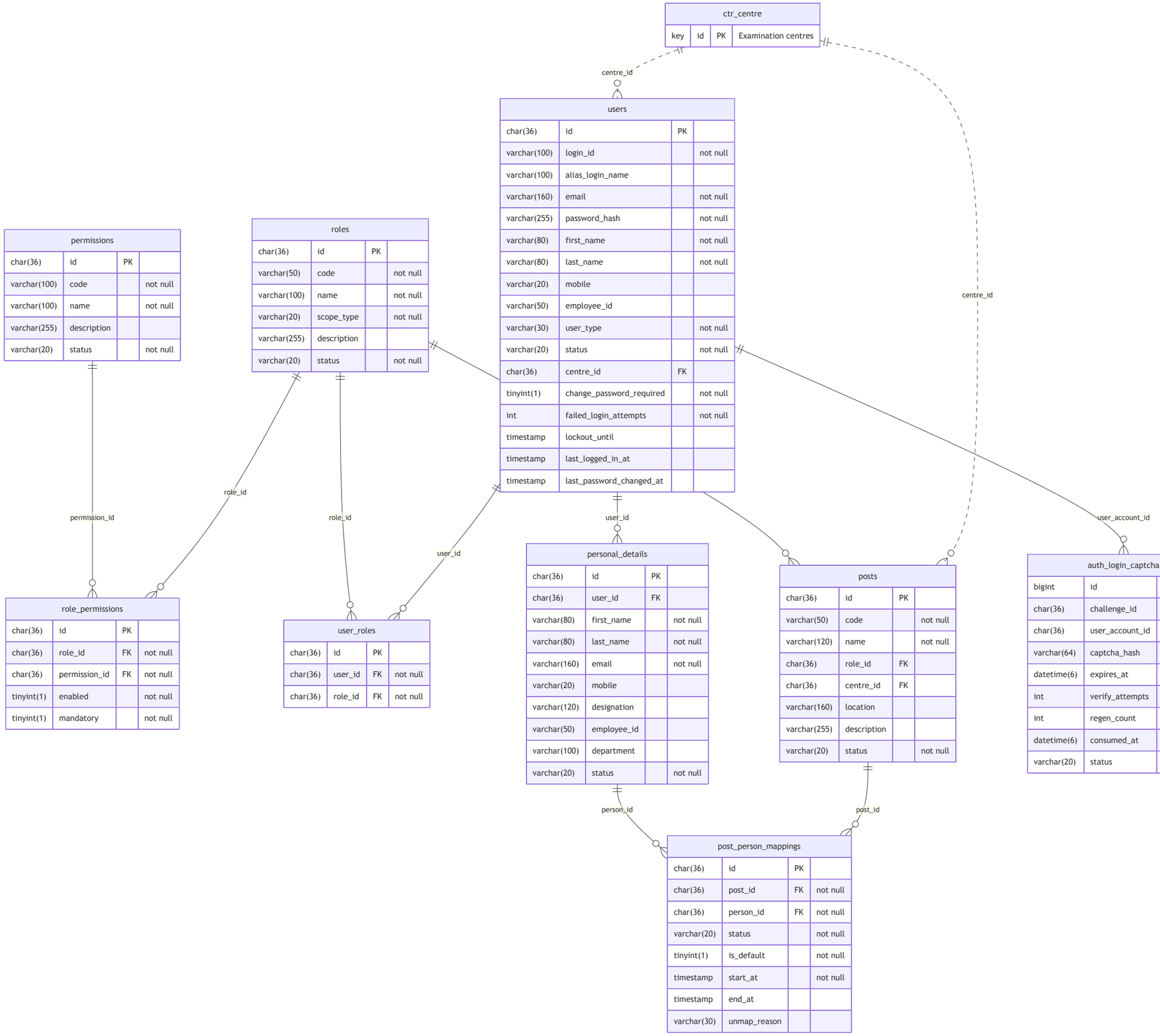
Database: MySQL (`exam_db`) · 56 tables · 48 foreign keys + 28 logical links · arranged by area

Legend: solid line = foreign-key constraint in the database; dashed line = logical link (ID stored, integrity enforced by the application, no database constraint). This page shows the areas and how many links run between them (arrow from the referring area to the referenced area). Each following page shows one area in full: links to tables in other areas are drawn to a box with that table's key and area name; links coming in from other areas are drawn on the page of the referring area. Standard audit columns (created_at, updated_at, created_by, updated_by and, where present, deleted_at, version) are omitted from the diagrams.



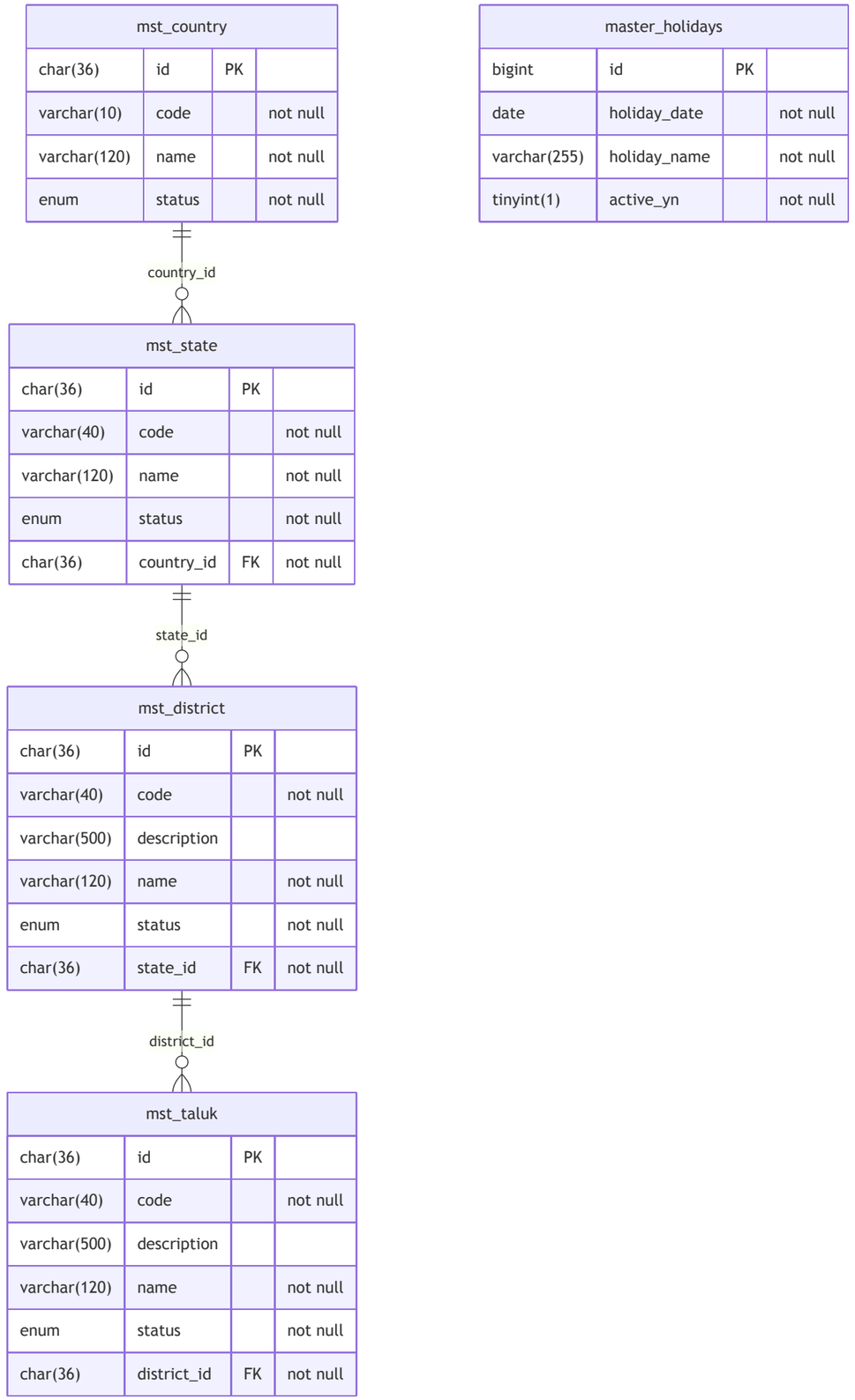
Users and roles

Tables: users, roles, permissions, role_permissions, user_roles, personal_details, posts, post_person_mappings, auth_login_captcha



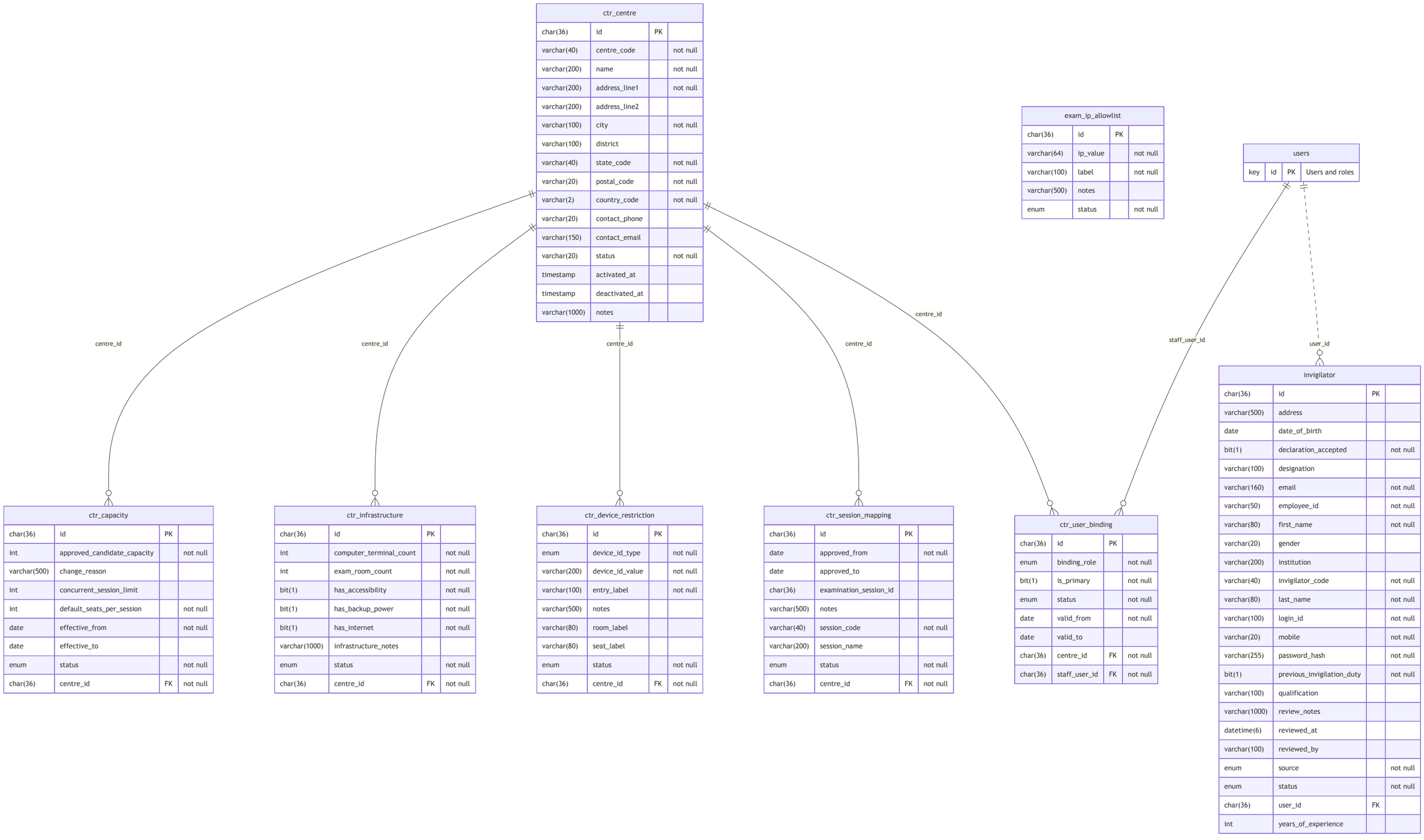
Masters and reference data

Tables: mst_country, mst_state, mst_district, mst_taluk, master_holidays



Examination centres

Tables: ctr_centre, ctr_capacity, ctr_infrastructure, ctr_device_restriction, ctr_session_mapping, ctr_user_binding, invigilator, exam_ip_allowlist



ctr_centre

char(36)	id	PK	
varchar(40)	centre_code		not null
varchar(200)	name		not null
varchar(200)	address_line1		not null
varchar(200)	address_line2		
varchar(100)	city		not null
varchar(100)	district		
varchar(40)	state_code		not null
varchar(20)	postal_code		not null
varchar(2)	country_code		not null
varchar(20)	contact_phone		
varchar(150)	contact_email		
varchar(20)	status		not null
timestamp	activated_at		
timestamp	deactivated_at		
varchar(1000)	notes		

ctr_capacity

char(36)	id	PK	
int	approved_candidate_capacity		not null
varchar(500)	change_reason		
int	concurrent_session_limit		
int	default_seats_per_session		not null
date	effective_from		not null
date	effective_to		
enum	status		not null
char(36)	centre_id	FK	not null

ctr_infrastructure

char(36)	id	PK	
int	computer_terminal_count		not null
int	exam_room_count		not null
bit(1)	has_accessibility		not null
bit(1)	has_backup_power		not null
bit(1)	has_internet		not null
varchar(1000)	infrastructure_notes		
enum	status		not null
char(36)	centre_id	FK	not null

ctr_device_restriction

char(36)	id	PK	
enum	device_id_type		not null
varchar(200)	device_id_value		not null
varchar(100)	entry_label		not null
varchar(500)	notes		
varchar(80)	room_label		
varchar(80)	seat_label		
enum	status		not null
char(36)	centre_id	FK	not null

ctr_session_mapping

char(36)	id	PK	
date	approved_from		not null
date	approved_to		
char(36)	examination_session_id		
varchar(500)	notes		
varchar(40)	session_code		not null
varchar(200)	session_name		
enum	status		not null
char(36)	centre_id	FK	not null

ctr_user_binding

char(36)	id	PK	
enum	binding_role		not null
bit(1)	is_primary		not null
enum	status		not null
date	valid_from		not null
date	valid_to		
char(36)	centre_id	FK	not null
char(36)	staff_user_id	FK	not null

exam_ip_allowlist

char(36)	id	PK	
varchar(64)	ip_value		not null
varchar(100)	label		not null
varchar(500)	notes		
enum	status		not null

users

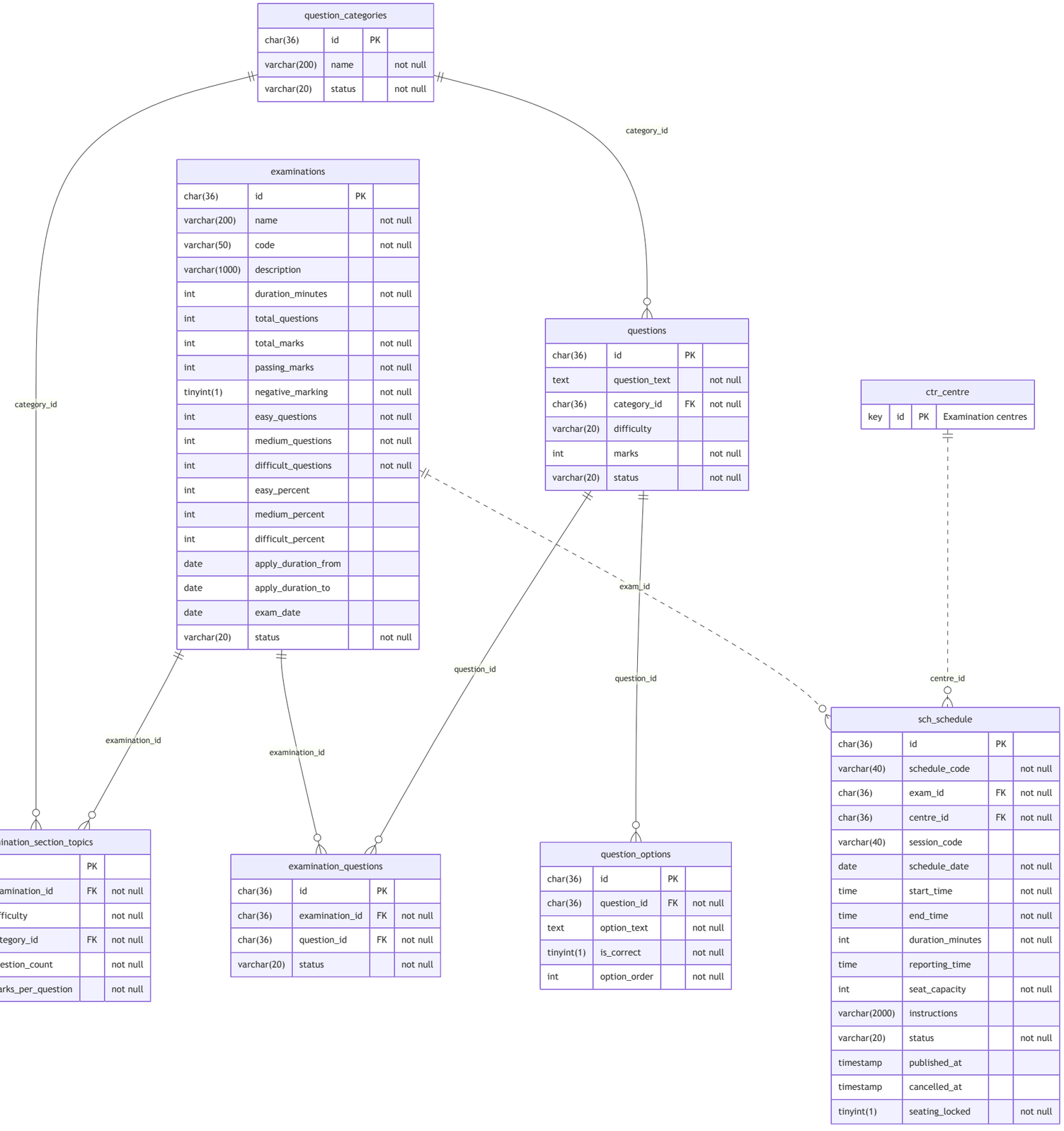
key	id	PK	Users and roles

invigilator

char(36)	id	PK	
varchar(500)	address		
date	date_of_birth		
bit(1)	declaration_accepted		not null
varchar(100)	designation		
varchar(160)	email		not null
varchar(50)	employee_id		not null
varchar(80)	first_name		not null
varchar(20)	gender		
varchar(200)	institution		
varchar(40)	invigilator_code		not null
varchar(80)	last_name		not null
varchar(100)	login_id		not null
varchar(20)	mobile		not null
varchar(255)	password_hash		not null
bit(1)	previous_invigilation_duty		not null
varchar(100)	qualification		
varchar(1000)	review_notes		
datetime(6)	reviewed_at		
varchar(100)	reviewed_by		
enum	source		not null
enum	status		not null
char(36)	user_id	FK	
int	years_of_experience		

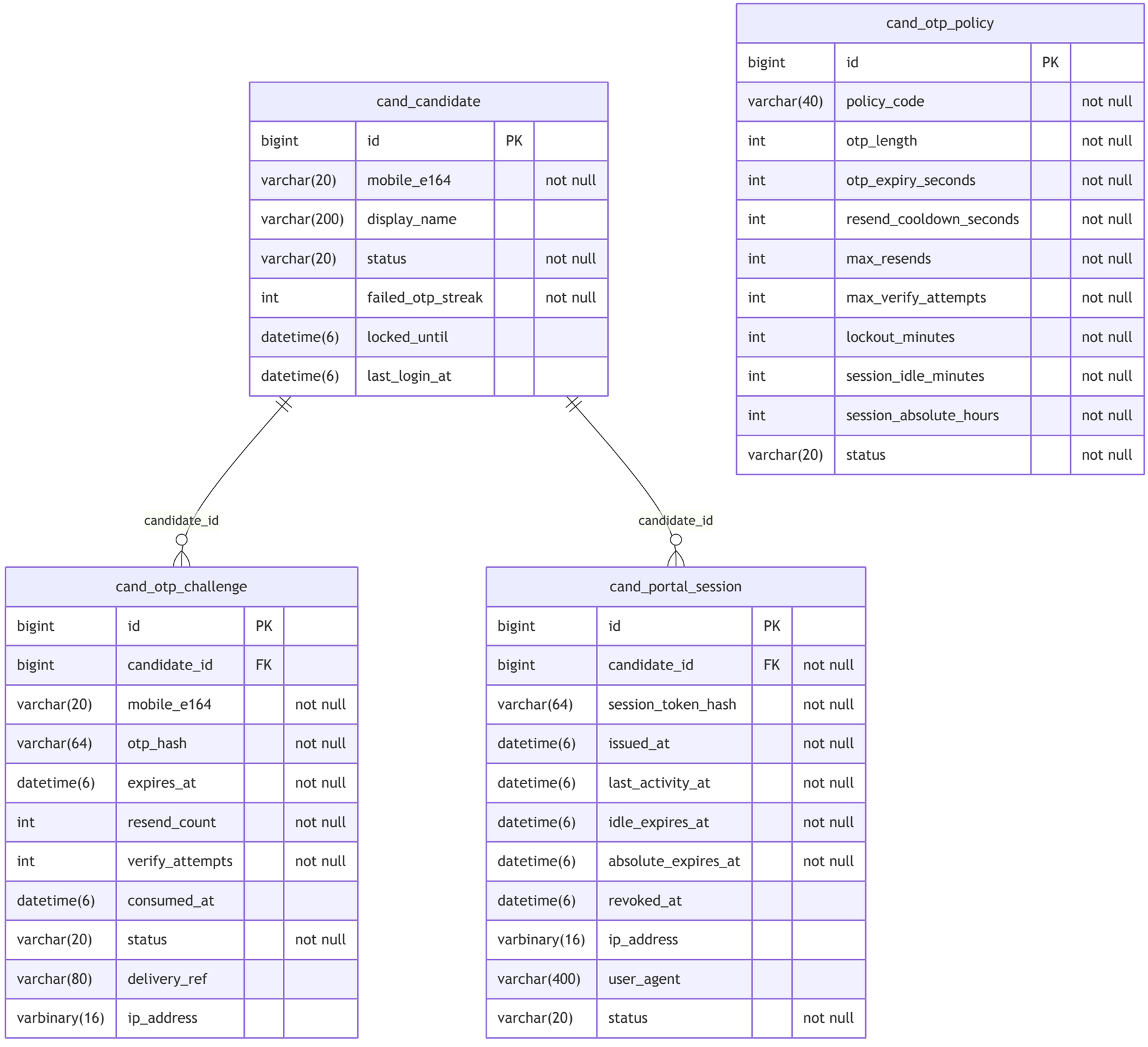
Examinations, question bank and schedules

Tables: examinations, examination_section_topics, examination_questions, questions, question_options, question_categories, sch_schedule



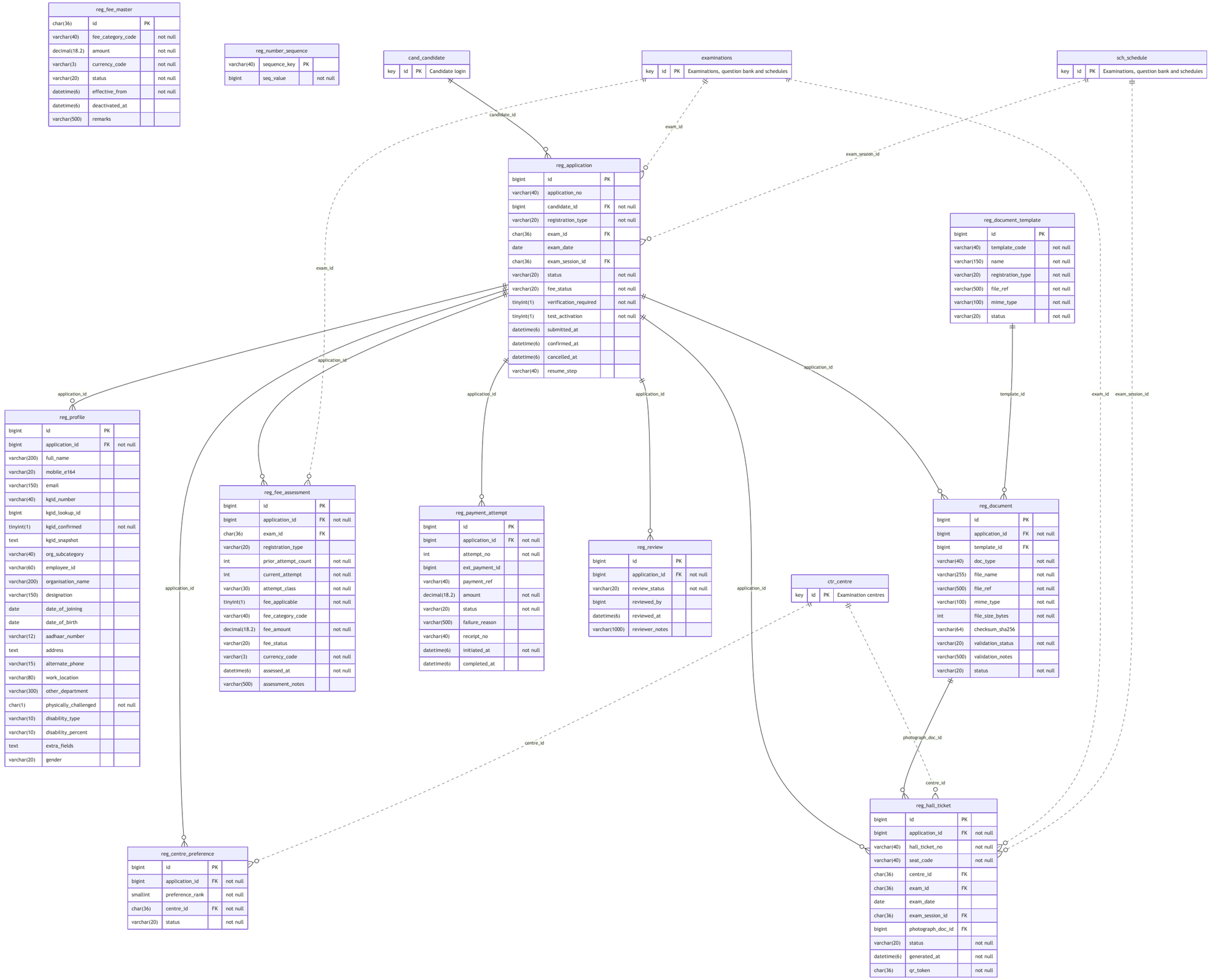
Candidate login

Tables: cand_candidate, cand_otp_challenge, cand_otp_policy, cand_portal_session



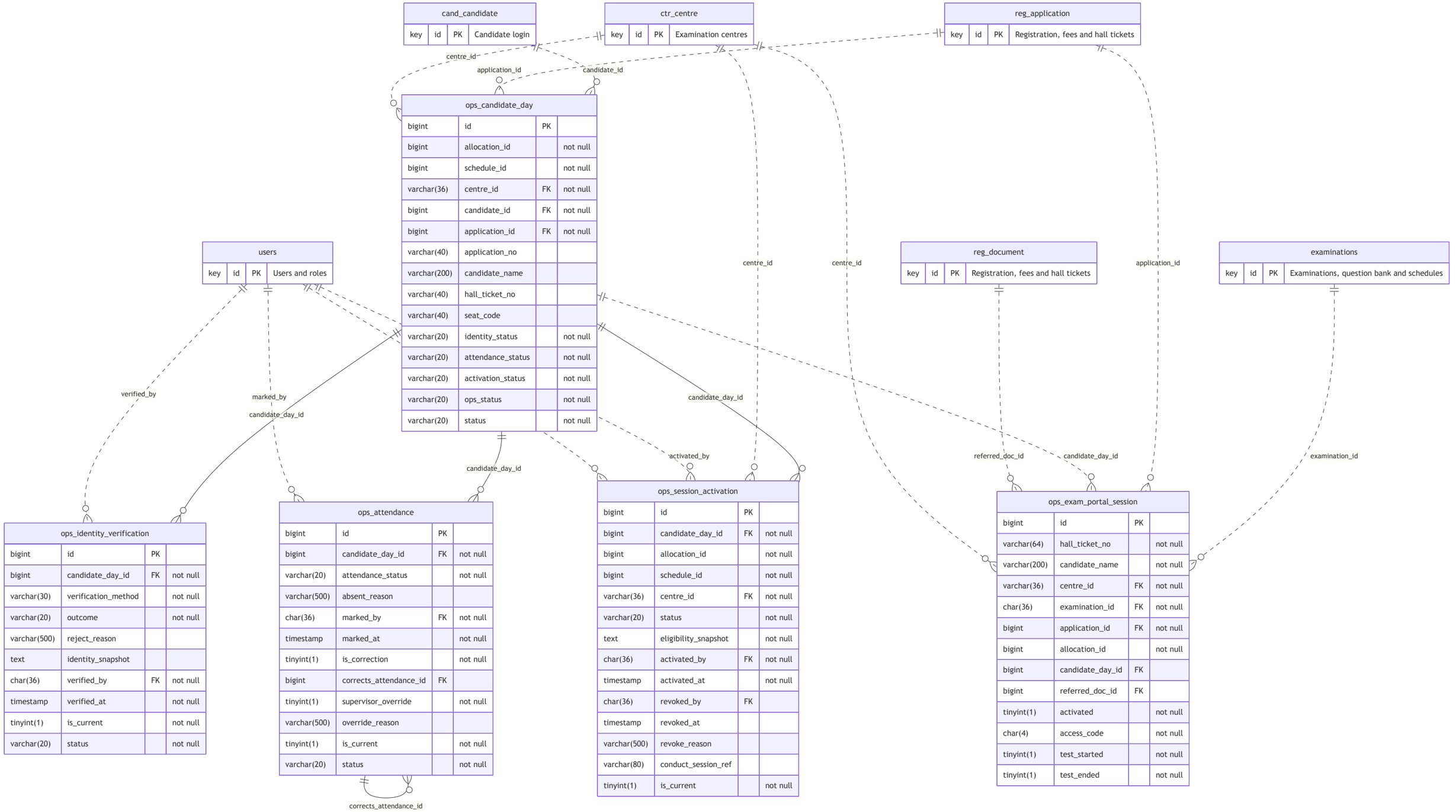
Registration, fees and hall tickets

Tables: reg_application, reg_profile, reg_centre_preference, reg_document, reg_document_template, reg_fee_master, reg_fee_assessment, reg_payment_attempt, reg_review, reg_hall_ticket, reg_number_sequence



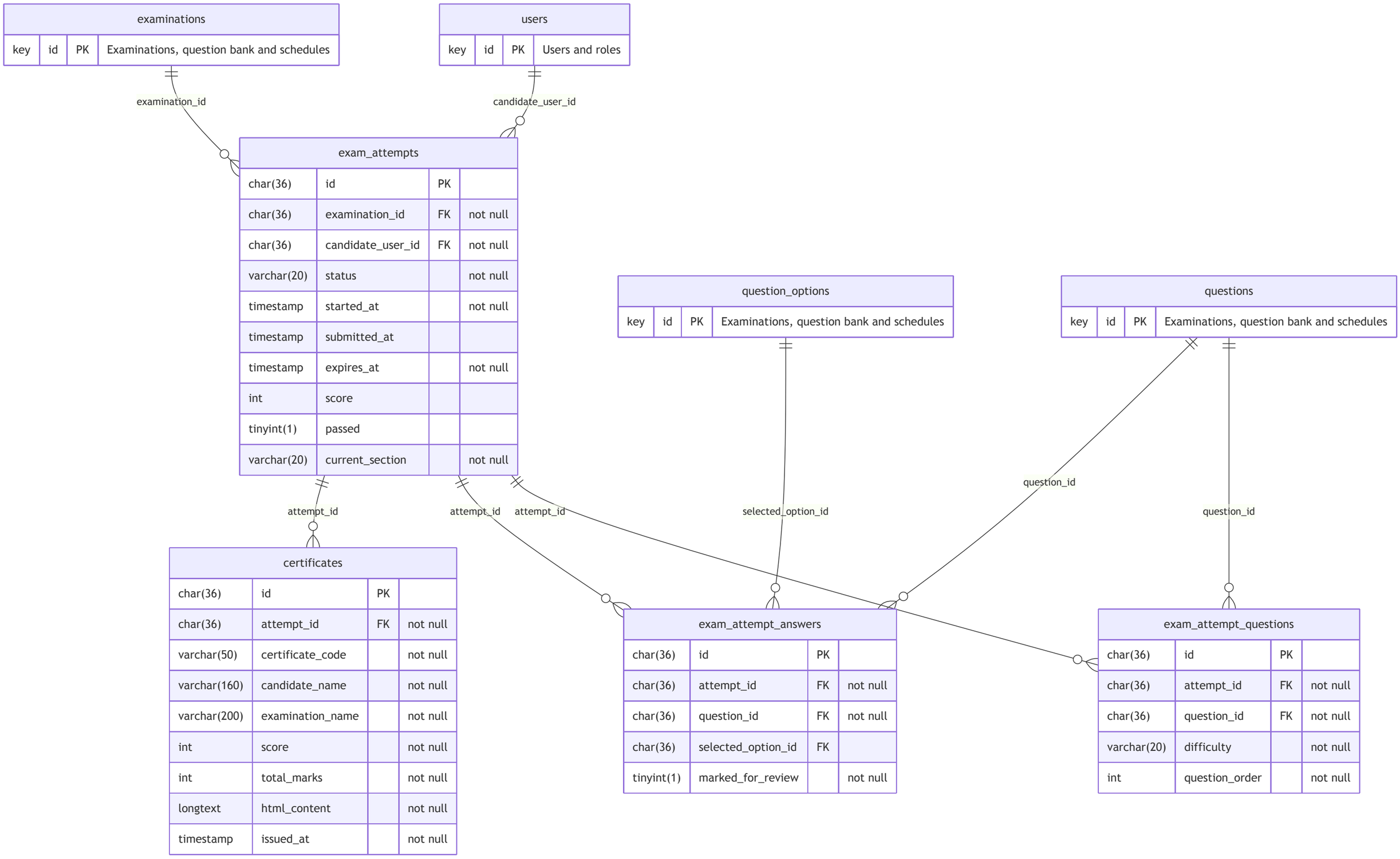
Exam-day centre operations

Tables: ops_candidate_day, ops_identity_verification, ops_attendance, ops_session_activation, ops_exam_portal_session



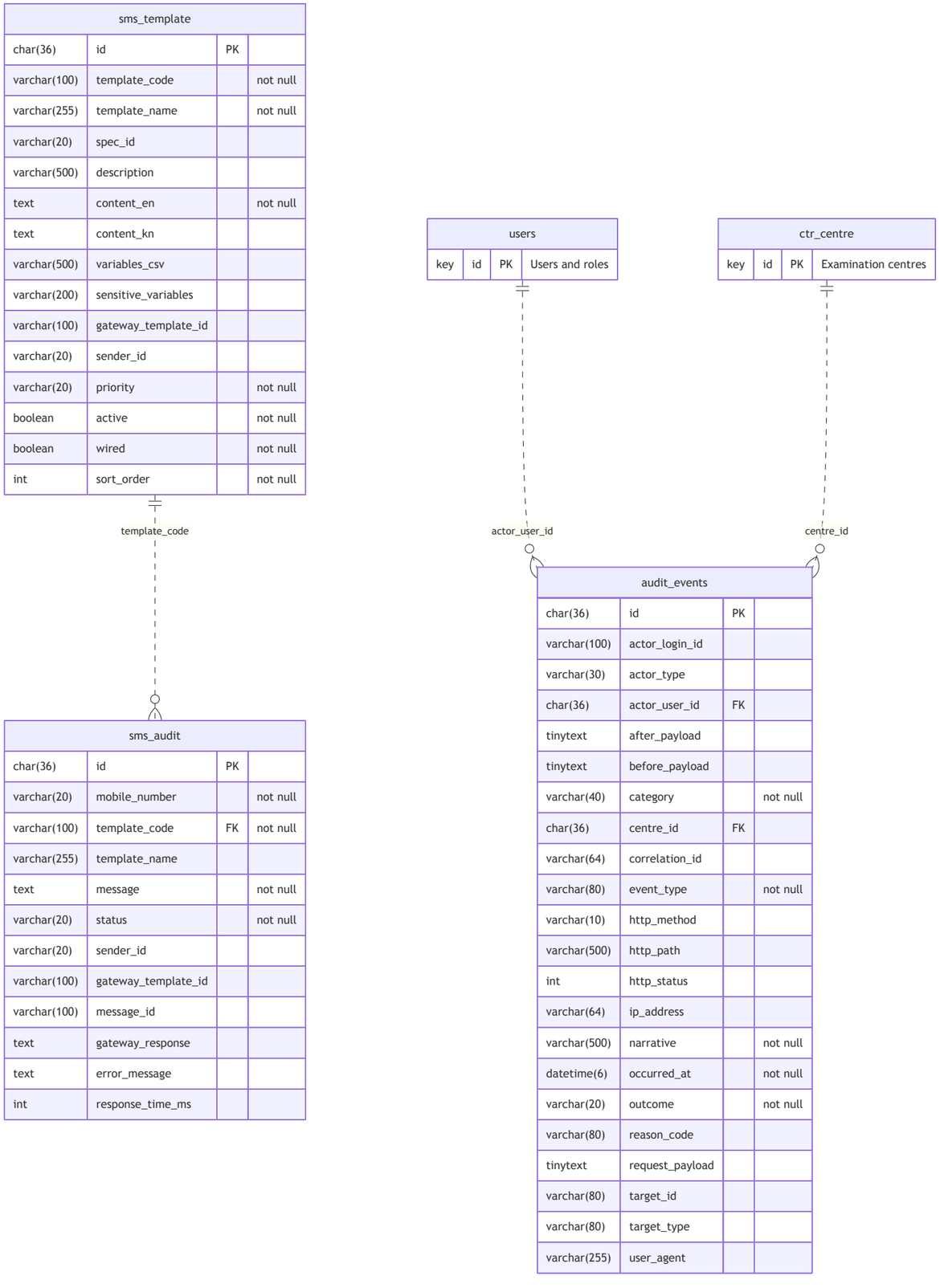
Online test and results

Tables: exam_attempts, exam_attempt_questions, exam_attempt_answers, certificates



Audit and notifications

Tables: audit_events, sms_template, sms_audit



Source

Structure taken from the `exam_db` schema (information_schema: columns, keys, foreign keys) and cross-checked against the JPA entity classes of the Examination Management Portal back end (branch `main`, commit `2a3186e`). All entity columns match the database.

`sms_audit` and `sms_template` are drawn from their entity classes; no Flyway migration creates them yet. `reg_fee_master` is drawn as defined by migration V42.

Logical links are ID columns that refer to another table without a database constraint (for example `reg_application.exam_id → examinations`, `reg_application.exam_session_id → sch_schedule`, `users.centre_id → ctr_centre`). `ops_candidate_day.schedule_id` holds a numeric key derived from the sitting and is therefore not drawn as a link.

Column-level details (data type, length, mandatory, default, description, validation rule, data classification) are in **ExamPortal_Data_Dictionary.xlsx**.